

Senior High School

Learning Activity Sheet for General Mathematics

Quarter 2

LAS

2

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Development Team

Writers: Elenita L. Nocidal
Jean L. Bataller
Richel L. Lozano
Juan A. Lappay
Mac Donald L. Dalugdog
Aljon S. Bucu, PhD
Lee C. Apas
Hernando C. Abalos, Jr

Validator: Romel C. Anaya
Jerson D. Jolo
Richard P. Ramos

Language Editor: Ferdinand Kevin Gumalay

Consultant: Dr. Flordeliza F. Francisco
Far Eastern University

Learning Area Specialist: Wilson R. Santiago

Bureau of Learning Delivery
Bureau of Curriculum Development
Bureau of Learning Resources

Every care has been taken to ensure the accuracy of the information provided in this material. For inquiries or feedback, please write or call the Office of the Director of the Bureau of Learning Resources via telephone numbers (02) 8634-1072 and 8631-6922 or by email at blr.od@deped.gov.ph

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	4.1	Date:	
Lesson Title/ Topic:	Quantities of Different Magnitudes: Choosing Appropriate Units and Applying Scientific Notation		
Name:		Grade & Section:	

I. Activity: Measure It Right: Math and Science in Everyday Life

II. Objectives:

1. Identify appropriate metric units for different real-life measurements
2. Convert numbers between standard form and scientific notation accurately.
3. Solve word problems involving large and small quantities using scientific notation.
4. Apply mathematical skills to interpret and analyze measurements used in everyday life.

III. Materials Needed: Calculator, Notebook or Writing Paper, Pencil or Pen

IV. Instructions:

Follow the directions for each part of the activity carefully. Make sure to write your answers clearly and show your solutions where needed.

Part 1: Identify the Appropriate Metric Unit

For each item, choose the most appropriate metric unit to measure what is being described. Think about whether the item requires measuring length, mass, volume, temperature, or time. Write the answer in the space provided.

1. measuring the length of a football field _____
2. weighing a newborn baby _____
3. determining the volume of a soft drink can _____
4. calculating the distance between two cities _____
5. timing a sprint race _____
6. measuring the Height of a Tree _____
7. measuring the temperature of a patient. _____
8. measuring the mass of a chemical compound _____

Part 2: Convert to Scientific Notation

Express each of the following numbers in scientific notation. Write the answer in the space provided.

1. 0.000451 _____
2. 78,000 _____
3. 56,000,000 _____
4. 123,000 _____
5. 0.00000032 _____
6. 0.0000071 _____
7. 0.000000000053 _____
8. 0.00000000379 _____

Part 3: Convert to Standard Notation

Rewrite each of the following numbers in standard form. Write the answer in the space provided.

1. 4.5×10^{-4} _____
2. 3.2×10^6 _____
3. 8.1×10^{-7} _____
4. 6.7×10^3 _____

5. 9.3×10^{-5} _____
6. 5.6×10^4 _____
7. 1.2×10^{-2} _____
- 7.8×10^8 _____

Part 4: Word Problems

Solve the following word problems involving scientific notation. Express your answers in scientific notation and include appropriate units.

1. Light travels at a speed of 3.00×10^8 meters per second. The distance from Earth to the Sun is approximately 1.496×10^8 kilometers. How many seconds does it take for light to travel from the Sun to Earth? Express your answer in scientific notation.
2. The diameter of the Sun is approximately 1.39×10^6 kilometers, while the diameter of Earth is about 1.27×10^4 kilometers. How many times larger is the diameter of the Sun compared to Earth? Express your answer in scientific notation.
3. The world population in 2025 is estimated to be 8.5 billion people. The population of a city is 4.2×10^6 people. What is the ratio of the city's population to the world population? Express your answer in scientific notation.
4. The Alpha Centauri system is approximately 4.37 light-years away from Earth. A spacecraft travels at a speed of 2.5×10^4 kilometers per second. If the spacecraft could travel directly to the Alpha Centauri system without stopping, how many seconds would it take to reach it? Express your answer in scientific notation. Note: 1 light-year $\approx 9.461 \times 10^{12}$ kilometers.

V. Synthesis:

1. How did you arrive at your answers?

2. How did you find the activity? Can you share and describe your experience?

3. How did using scientific notation make it easier to solve problems with really large or small numbers? Can you think of a real-life situation where this would be helpful?

4. What's the most important thing you learned from this activity about scientific notation and measuring things?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	4.2	Date:	
Lesson Title/ Topic:	Unit Conversion Within and Across Systems		
Name:		Grade & Section:	

I. Activity: Conversion Escape Room

II. Objectives:

Convert units accurately between metric and English measurement system.

Apply unit conversions to solve real-life, scenario-based problems.

Demonstrate critical thinking and teamwork in completing multi-step math challenges.

III. Materials Needed: Printed activity sheets, Pencils or pens, Calculator,

IV. Instructions:

Mechanics:

1. Form a Team

(Or play individually, depending on your teacher's instruction.)

2. Go Through Each Station

There are **3 stations**, each with a different conversion challenge:

Station 1: The Water Lock

Station 2: Temperature Code

Station 3: Volume Vault

3. Solve the Challenge

Write your complete answer for each station in the space provided.

4. Extract the Code

From each correct answer, take the **first digit only** (example: for 2.5, record "2").

Unlock the Vault

Combine the 3 digits from all stations to form your **Final Escape Code**.

Check Your Code

If your code is **correct**, you've successfully escaped!

Complete the Reflection

After escaping, answer the short reflection questions as part of your learning check.

5. Mission:

Your team is trapped in a vault! To escape, you must complete three challenges involving unit conversions — both within and between metric and English units. Solve them all to unlock the final code!

Station 1: The Water Lock

💧 You need 2,500 milliliters of a special chemical. Your container only shows liters.

Challenge: How many liters is 2,500 milliliters?

✍ Answer: _____

🔑 Clue Code (first digit of your answer): _____

Station 2: Temperature Code

🌡 The climate control panel is set in Celsius, but the machine only reads Fahrenheit.

Challenge: Convert 30°C to degrees Fahrenheit.

✍ Answer: _____

🔑 Clue Code (first digit of your answer): _____

Station 3: Volume Vault

📦 The machine must be filled with exactly four quarts, but your tools only measure in cups.
Challenge: How many cups is four quarts?

✎ Answer: _____

🔑 Clue Code (first digit of your answer): _____

Final Escape Code:

Combine the first digits from each station below:

➡ Code: _____ - _____ - _____

🚪 If your code is correct, you've successfully escaped the vault!

🗨 Victory Reflection:

Take a few minutes to answer:

1. Which challenge was the easiest or hardest? Why?

2. How did you approach the conversions?

3. Give 1 real-life example where unit conversion is useful.

V. Synthesis:

1. What did you learn about unit conversions during the activity?

2. Which station was the easiest or most challenging for you? Why?

3. How did your team work together to solve the challenges (if done in groups)?

4. What strategies helped you solve the problems faster or more accurately?

5. If you were to design your own escape room challenge, what kind of conversion task would you include?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	4.3	Date:	
Lesson Title/ Topic:	Determining the surface area of three-dimensional objects such as prisms, pyramids, cylinders, cones, spheres and combinations		
Name:		Grade & Section:	

I. Activity No. 1: Surface Area Stations

II. Objective: At the end of the activity, you should be able to compute surface area using given dimensions at each station.

III. Materials Needed: pen and paper, measuring device (ruler, tape measure), calculators

IV. Instructions:

1. Form groups of five to six members. Each group will be assigned by your teacher to visit **three different stations**, each with a different solid shape.
2. Check if the station assigned to your group is available. If another group is already working at your assigned station, wait for your turn. Please be patient and respectful of other groups.
3. At each station, you will see: a picture or model of the shape, its dimensions (lengths, radius, height, etc.), a measuring device and a surface area problem card with a given scenario and task.
4. Read each scenario and task carefully, then discuss with your group members how to approach the solution. Use a measuring device (if needed) to determine the dimensions of the 3D shapes. Use the correct formula (from the formula bank) to compute the surface area.
5. Write your solutions and final answers on your **station worksheet**.
6. After three minutes, move to the next assigned station when your teacher signals.

Important Reminders:

- Make sure to show your work and calculations for each problem.
- Use the formula bank and calculators provided, if needed.
- If you have questions or if you need help, don't hesitate to ask your teacher.
- Stay focused and work together as a team.
- Be mindful of the time and manage it effectively to complete the solutions to the problems at each assigned station.
- Have fun and enjoy the challenge.

Station 1: "Gift Wrapping a Toy Box"

Scenario: You are wrapping a toy box as a birthday present. The box is shaped like a rectangular prism.

Task: 1. Measure the dimensions of the toy box.

Calculate the total surface area of the toy box to know how much wrapping paper you need.

Station 2: "Designing a Soda Can Label"

Scenario: You are helping design a new label for a soda can. The label must cover the entire surface area of the can (including the top and bottom).

Task: 1. Measure the radius and height of the soda can.

2. Find the total surface area of the cylinder so you can determine the correct label size.

Station 3: "Designing a Soda Can Label"

Scenario: You are helping design a new label for a soda can. The label must cover the entire surface area of the can (including the top and bottom).

Task:

1. Measure the radius and height of soda can.
2. Find the total surface area of the cylinder so you can determine the correct label size.

Station 4: Sphere – "Painting a Ball"

Scenario: You are repainting a smooth decorative ball of radius 6 cm for a school project. You need to know how much paint will be required.

Task: Compute the surface area of the sphere to find how much surface needs to be painted.

Station 5: Square-Based Pyramid – "Designing a Tent"

Scenario: You are designing a tent shaped like a square-based pyramid. The surface includes the square base and four triangular faces.

Base side length: 5 m

Slant height: 8 m

Task: Calculate the total surface area of the pyramid to know how much fabric is needed.

Station Worksheet

Station #	3D Shape	Dimensions	Surface Area Formula	Computation for Surface Area

Formula Bank

(Use each formula only once. Match it with the correct 3D shape above.)

- A. $SA = 6s^2$
- B. $SA = 2lw + 2lh + 2wh$
- C. $SA = 2\pi r^2 + 2\pi rh$
- D. $SA = \pi r^2 + \pi rl$
- E. $SA = 4\pi r^2$
- F. $SA = b^2 + 2bl$ (where b = base, l = slant height)

V. Synthesis:

1. Compare and contrast the process of finding the surface area of a rectangular prism and a cylinder. What similarities and differences did you notice in the steps and formulas used?

2. In real-life scenarios, why is calculating surface area important?

3. Which station did you find most challenging and why? Was it due to the shape, formula, measurement, or computation? How did you overcome this challenge?

4. If the surface area is crucial in designing or packaging, what could go wrong if surface area is miscalculated? Relate your answer to one of the scenarios in the stations.

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	4.4	Date:	
Lesson Title/ Topic:	Calculating the Volume of Prisms, Pyramids, Cylinders, Cones, Spheres and Combinations of these.		
Name:		Grade & Section:	

I. Activity No. 1: Volume Drill & Skill Stations

II. Objectives: At the end of the activity, you should be able to calculate the volume of 3D shapes (prisms, pyramids, cylinders, cones, spheres and combinations of these).

III. Materials Needed:

pen and paper, notebook

IV. Instructions:

1. Form groups of five to six members. Each group will be assigned by your teacher to visit the five different stations, with tasks involving different 3D shapes.
2. Check if the station assigned to your group is available. If another group is already working at your assigned station, wait for your turn. Please be patient and respectful of other groups.
3. Groups need to take turns as they rotate through the 5 stations.
4. At each station, each group will:
5. read the problem involving the shape.
6. use the correct formula to compute volume.
7. show complete solutions in a piece of paper or notebook.
8. Work collaboratively with your group members. Share your ideas, discuss any challenges, and help each other out.
9. After 3 minutes, move to the next station when your teacher signals.

Finish the Challenge Task (Bonus). At this final station, solve a mixed-shape word problem that involves finding the total volume of a composite solid.

Each group must let the teacher check the answers after completing the tasks.

Important Reminders:

- Make sure to show your work and calculations for each problem.
- If you have questions or if you need help, don't hesitate to ask your teacher.
- Stay focused and work together as a team.
- Be mindful of the time and manage it effectively to complete the solutions to the problems at each assigned station.

Station 1: Rectangular & Triangular Prisms

Focus: Apply and interpret volume formula for prisms

Formula:

Rectangular Prism: $V = l \times w \times h$

Triangular Prism: $V = \frac{1}{2} \times b \times h \times l$

Tasks:

A box has a length of 10 cm, width of 6 cm, and height of 8 cm.

► What is its volume?

Challenge Problem (Application):

A water tank in the shape of a triangular prism has a base of 4 m, height of triangle 3 m, and length 5 m.

► How many cubic meters of water can it hold when full?

Station 2: Cylinders

Focus: Master volume of cylinders, circular base area application

Formula: $V = \pi r^2 h$

Tasks:

A cylindrical can has a radius of 5 cm and a height of 12 cm.

► What is its volume? (Use $\pi = 3.14$)

Visual Reasoning:

Suppose you double the radius but keep the height the same.

► How does the volume change? Explain briefly.

Station 3: Cones

Focus: Use and compare with cylinder formula.

Formula: $V = \frac{1}{3} \pi r^2 h$

Tasks:

A cone has a radius of 6 cm and a height of 9 cm.

► Find the volume of the cone. (Use $\pi = 3.14$)

Compare with a Cylinder:

A cone and a cylinder have the same base radius and height.

► What is the relationship between their volumes?

Station 4: Pyramids

Focus: Volume of square or rectangular pyramids

Formula: $V = \frac{1}{3} \times \text{Base Area} \times h$

Tasks:

A square pyramid has a base of 10 m $\times$ 10 m and a height of 15 m.

► What is its volume?

Word Problem:

A sandbox is shaped like a rectangular pyramid with base 6 m by 4 m and height 3 m.

► How much sand (in m^3) will fill it to the top?

Station 5: Spheres

Focus: Conceptual and computational fluency

Formula: $V = \frac{4}{3} \pi r^3$

Tasks:

A basketball has a radius of 12 cm.

► What is its volume? (Use $\pi = 3.14$)

Application Challenge:

A spherical balloon expands from radius 2 cm to 4 cm.

► By what factor does the volume increase?

Bonus Challenge Station: Mixed Solids

Focus: Real-life applications and problem-solving.

Task:

A candle is shaped like a cylinder (radius 3 cm, height 10 cm) with a conical top (same radius, height 5 cm).

► What is the total volume of the candle?

Synthesis:

1. Across all stations, what did you notice about how the base area influences the volume of a 3D shape? Use examples from at least one of the stations to support your answer.

2. How are the volume formulas for prisms and pyramids similar and different?

3. Compare the volume formulas of a cylinder and a cone.

4. Reflect on the relationship between radius and volume. Using the sphere and cylinder stations, how does changing the radius affect the total volume of a shape?

5. Think about the bonus station problem. How is finding the volume of a composite solid (cylinder + cone) different from the problems at the earlier stations? What strategies helped you break it down?

6. Which station made you think the most, and why? Was it due to formula complexity, shape visualization, or real-life context?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	4.5	Date:	
Lesson Title/ Topic:	Converting Currencies and Time, and Using Maps in Traveling Decision		
Name:		Grade & Section:	

I. Activity No. 1: “Budgeting and Time Travel Across Borders”

II. Objective: At the end of the activity, you should be able to convert currencies and time zones across countries.

III. Materials Needed:

Currency Exchange Table (printed or online, time zone chart or access to time zone converter, calculator or smart phone (optional)

IV. Instructions:

 Analyze and solve each problem below.

1. Nicole is traveling from the United States to Europe. She has 500 USD and wants to convert it into Euros (EUR). The current exchange rate is 1 USD = 0.88 EUR. How many Euros will Nicole have?

2. Mr. De La Cruz prepares his trip from the Philippines to the United States. He needs to exchange Philippine pesos (P) for US dollars (USD). With 100,000.00 Philippine pesos, he wants to find out how much US dollars he will get based on the exchange rate of 1 USD = 55.42. How much in US dollars will Mr. De La Cruz get?

3. Katrina wants to buy a dress from an online store in the US, where the price is 150 USD. The exchange rate is 1 USD = 55.57 P. How can Katrina convert the dress price to Philippine pesos?

4. Ariel lives in Albay, Philippines and he has a virtual job interview scheduled with a company based in New York City, USA. The interview is scheduled for 10:00 AM Eastern Time (ET) on December 10, 2025. What time and date should Ariel log in for the interview in Philippine Time (PHT)?

5. Lea is from Makati City, Philippines. She wants to call her mother who works in Dubai, United Arab Emirates (Gulf Standard Time – GST). She is free to call at 7:00 PM Philippine Time after doing her homework. What time will it be in Dubai when she makes the call?

5. Criselda, an OFW nurse in Seoul, South Korea (Korean Standard Time – KST), wants to call her children in Bangkok, Thailand (Indochina Time – ICT). Criselda will call at 8:00 PM KST after her shift in the hospital where she works. What time will it be in Thailand by then?

V. Synthesis:

1. How do we convert currencies?

2. Why is it important to know how to convert currencies, especially when traveling?

3. What are the steps in converting time between countries?

4. In what ways can time zone differences affect international communication, like scheduling online classes, meetings, or flights?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	5.1	Date:	
Lesson Title/ Topic:	Functions		
Name:		Grade & Section:	

I. Activity No. 1: Understanding Functions

II. Objective: At the end of the activity, you should be able to determine measures of central tendency of ungrouped data and use the measures to draw conclusions.

III. Materials Needed: pen and paper

IV. Instructions: Analyze and solve each problem.

1. If the volume V of a cube is related to the side length s by $V=s^3$, how can you determine s given V ? Which function would you use? Given the Volume V is 25 cm^3 , what is the side length s ?
2. In a small coastal town in Zamboanga, Aling Nena runs a sari-sari store that sells local brands of *sardinas*. One of the best-selling items is a cylindrical tin of sardines from a brand called “Masarap Sardines”. The label on the can says it contains Volume: 350 cm^3 of sardines. Aling Nena measures the height of the can and finds it to be 10 cm. She wants to stack the cans efficiently on her shelf and needs to know the radius of the base of each can, to calculate how much space she needs.
3. Miguel, a Grade 11 student from Cebu, wants to surprise his younger sister with a gift. He chooses a small box shaped like a cube, which he plans to wrap with colorful gift wrapper bought from the nearby market. The box has a volume of 64 cm^3 , and Miguel wants to know how much wrapping paper he needs to cover the entire surface of the box without leaving any side exposed.

V. Synthesis:

1. How can we use volume formulas to find missing parts of 3D objects like cubes and cylinders in real-life situations?

2. Why is it important to understand the shape and size of objects (like cans or boxes) when solving problems at home, in stores, or when giving gifts?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	5.2	Date:	
Lesson Title/ Topic:	Linear and Quadratic Functions		
Name:		Grade & Section:	

I. Activity: Understanding the Linear and Quadratic Functions

II. Objective: At the end of the activity, you should be able to demonstrate prior knowledge and understanding of linear and quadratic functions by accurately identifying key characteristics, forms, and applications

III. Materials Needed: pen and paper, graphing paper

IV. Instructions:

Activity:

Choose the letter of the correct answer.

- Which of the following best defines a function?
 - A relation where each input has exactly one output
 - A relation where inputs can have multiple outputs
 - A set of ordered pairs
 - A graph that forms a straight line
- What is the slope of the linear function $f(x) = 5x - 2$?
 - 2
 - 5
 - 2
 - 5
- Which of the following represents a quadratic function?
 - $y = 3x + 5$
 - $y = 3x^2 - 2x + 1$
 - $y = \frac{1}{x}$
 - $y = x^3$
- What is the shape of the graph of a quadratic function?
 - Line
 - Circle
 - Parabola
 - Triangle
- What is the y-intercept of the line $y = -2x + 5$?
 - 2
 - 5
 - 5
 - 2

Activity 2: Quick Plot Challenge

Instructions:

Get Ready: Grab your graphing paper, ruler, and pencil.

Graph These Functions:

$$y = 2x - 1$$

$$y = x^2 + 2$$

Make a Table:

Choose 5 values for x (like -2, -1, 0, 1, 2).
Find the matching y values for each function.
Plot the points on your graph.

Label Everything:

Write the name of each function as the title.
Label the x -axis and y -axis.

Mark at least 3 important points (like where the graph crosses the axes or the highest/lowest point).

Activity 3 : Evaluating Learning

Solve the following:

1. Problem: A tricycle driver charges a base fare of ₱10 plus ₱5 per kilometer traveled.

Question: Write a linear function that models the total fare F for traveling x kilometers. How much would it cost to travel 8 kilometers?

2. Problem: A local festival in Cebu features a fireworks display. The height h of a firework in meters after t seconds is given by the function

$$h(t) = -4.9t^2 + 29.4t + 1.$$

Question: How long does it take for the firework to reach its maximum height? What is the maximum height

3. Problem: Revenue Optimization for a Local Concert

A student organization in **Cebu City** is organizing a benefit concert. Tickets are priced at **₱300**, and they expect **200 attendees** at that price. They estimate that for every **₱20 increase** in ticket price, **10 fewer people** will attend.

Write a quadratic function that models the **total revenue** based on the number of ₱20 price increases.

Determine the ticket price that will **maximize revenue**.

Find the **maximum possible revenue**.

Interpret the result in the context of the problem.

Activity 4: Performance Task: Quadratic Functions in Real Life

Title: "Profit Planning for a Local Business"

Objective:

Apply your understanding of quadratic functions to model and solve a real-world problem involving profit optimization.

Task Description:

You are a young entrepreneur in the Philippines planning to sell a local product (e.g., halo-halo, tote bags, t-shirts, or snacks). You want to determine the best price to **maximize your profit**.

You will:

Create a quadratic function that models your profit based on price changes.

Convert the function to vertex form to find the price that gives maximum profit.

Graph the function and interpret the results.

Present your findings in a short report or presentation.

Guidelines:

Choose a product and set a base price and expected number of customers.

Assume that for every price increase, a certain number of customers stop buying.

Let x be the number of price increases.

Write the **profit function**:

$P(x) = (\text{price per item})(\text{number of customers})$

Expand and simplify the function into standard form.

Convert it to **vertex form** and identify the **maximum profit** and **optimal price**.

Graph the function and label the vertex.

Write a **1-page reflection** explaining:

Your process

What the vertex means

How this applies to real business decisions

Rubric

Criteria	Points
Correct quadratic model	4 pts
Accurate conversion to vertex form	4 pts
Graph with labeled vertex	4 pts
Realistic scenario and interpretation	4 pts
Clarity and completeness of reflection	4 pts
Total	20 points

Synthesis:

1. How did you arrive in solving problems involving linear and quadratic functions?

2. Give one real-life situation that can be modeled by a linear function.

3. Give one real-life situation that can be modeled by a quadratic function.

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	5.3	Date:	
Lesson Title/ Topic:	Piecewise Function		
Name:		Grade & Section:	

I. Activity No. 1: Delving Deeper

II. Objectives: At the end of the activity, you should be able to:

1. graph and represent piecewise function in practical context,
2. solve practical problems involving piecewise functions

III. Materials Needed: pen and paper

IV. Instructions: Read and understand the following problems. Solve each item. Sketch the graph and find the following:

$$f(x) = \begin{cases} 2x + 5, & x \leq -1 \\ 8 - x, & x > -1 \end{cases}$$

$f(-8)$
 $f(-1)$

c. $f(9)$
d. $f(12)$

$$g(x) = \begin{cases} 3, & x < -3 \\ 8 + x, & -3 \leq x \leq 6 \\ 4x, & x > 6 \end{cases}$$

$g(-9)$
 $g(-3)$

c. $g(0)$
d. $g(4)$

e. $g(6)$
f. $g(12)$

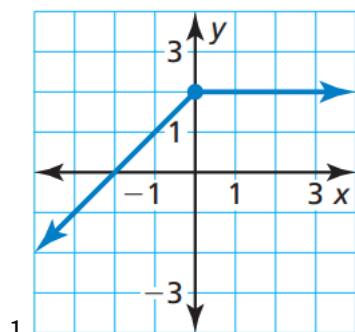
$$h(x) = \begin{cases} x^2 + 5, & -4 < x < 1 \\ 5, & 1 \leq x \leq 4 \\ 9 - x, & 4 < x \leq 12 \end{cases}$$

$h(-12)$
 $g(-4)$

c. $g(0)$
d. $g(4)$

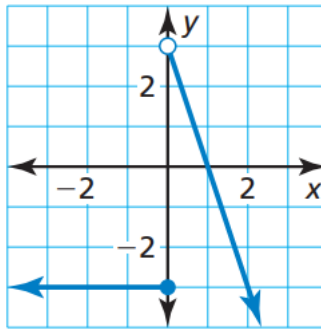
e. $g(8)$
f. $g(12)$

Write a piecewise function for each indicated condition. Write your answer on the blank.



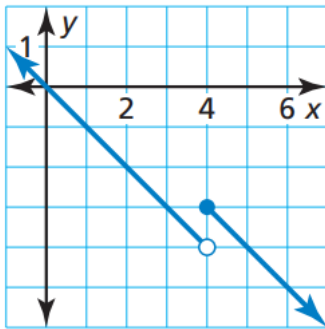
1

Find:
1. $f(-1)$
2. $f(0)$
3. $f(1)$



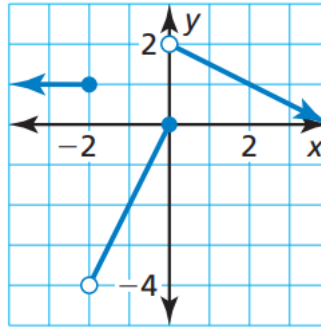
2

Find:
1. $f(-2)$
2. $f(0)$
3. $f(1)$



Find:
 1. $f(2)$
 2. $f(4)$
 3. $f(6)$

3 _____ 4



Find:
 1. $f(-2)$
 2. $f(-1)$
 3. $f(0)$
 4. $f(1)$

5. $f(x) = 3|2x - 7|$ _____

6. $f(x) = 2|x - 1| + 3$ _____

7. Taxi Fare

Base fare: ₱35 for the first 1.5 km

₱8 per km for the next 10.5 km

₱20 per km beyond 15.5 km

8. Overtime Pays

Regular pay: ₱80/hour for up to 6 hours

Overtime: ₱100/hour beyond 6 hours

9. Parking Fees

P100 for the first hour

P75 for each additional hour up to 5 hours

Flat P500 for more than 5 hours

10. Water Bills

First 10 m³: ₱20/m³

Next 20 m³: ₱30/m³

Beyond 30 m³: ₱50/m³

Activity No. 2: Deeping the Core of Understanding

A certain Liquor is sold for P150.00 each. With an increasing public demand, a vendor decides to sell it for P140.00 each if someone buys more than 15 bottles.

Express the cost T with respect to the number x of bottled liquors sold.

Calculate the $T(8), T(15), T(20)$.

Draw the graph.

Rental cars charge P100.00 for the first three hours and an excess of P20.00 for each hour (or a fraction of it) after that. If you rent a car for more than ten hours, a fee of P500.00 shall be charged.

Represent the rental car fee in piecewise function

How much will be paid if the rental time are as follows (a). 1.8 hrs., (b) 4.2 hrs., (c) 12.7 hrs.

A tricycle ride costs ₱10.00 for the first 3 kilometers, and each additional full kilometer adds ₱2.00 to the fare.

Let d be the distance (in kilometers) of the ride.

Use a piecewise function to represent the total fare $f(d)$ in terms of d .

Find the total fare $f(d)$ if $d = 1.8, 2.1, 3.4, 4.8$.

A company offers two different pay structures for its sales representatives:

Plan A: A fixed base salary of **P5,000** with **no commission** for sales below **P10,000**. If sales reach **P10,000 or more**, the salesperson earns **P5,000 plus a 10% commission on all sales**.

Plan B: A **lower base salary of P3,000** but a **15% commission on all sales**, regardless of the amount.

Define the total earnings function $f(x)$ for **Plan A** and $g(x)$ for **Plan B**.

Questions to Solve:

Write the piecewise function for $f(x)$ (Plan A) and the linear function for $g(x)$ (Plan B).

Determine which plan provides better pay for **sales of P8,000**.

Determine which plan provides better pay for **sales of P15,000**.

When is Plan A better than Plan B?

Activity 3. Performance Task: Investigating Electricity Billing

Scenario:

Electricity bills in the Philippines are often based on a tiered pricing system, where the cost per kilowatt-hour (kWh) increases with higher usage. However, the exact structure is not always clear to consumers.

Your Task:

You are an energy analyst tasked with understanding how electricity billing works in your area. Your goal is to **model the billing system using a piecewise function**.

Steps:

Research or collect data from actual electricity bills (your own, family members, or sample bills online). Identify how the total bill changes with electricity usage (in kWh).

Analyze the data to determine if there are different pricing tiers (e.g., a base rate, then higher rates for higher usage).

Define a piecewise function ($E(x)$) that models the total cost of electricity based on usage (x) in kWh.

Graph the function to visualize how the cost increases with usage.

Reflect and discuss:

What strategies could help reduce electricity costs?

Present your findings in a short report or presentation.

Rubric: Modeling Electricity Billing with Piecewise Functions (AI generated but modified)

Criteria	Excellent (4 pts.)	Good (3 pts.)	Satisfactory (2 pts.)	Needs Improvement (1 pt.)
Data Collection & Analysis	Collected accurate and relevant data; clearly identified pricing tiers	Collected data and identified most pricing tiers	Collected some data; pricing tiers unclear or incomplete	Little or no data collected; no clear analysis
Piecewise Function	Function is accurate, complete, and clearly written	Function is mostly accurate with minor errors	Function has some errors or missing parts	Function is incorrect or missing
Graph	Graph is accurate, labeled, and clearly shows piecewise behavior	Graph is mostly accurate with minor issues	Graph is present but has major errors or is unclear	Graph is missing or incorrect
Reflection & Discussion	Insightful analysis of fairness and impact; well-supported	Good analysis with some supporting points	Basic reflection with limited support	Little or no reflection or analysis

Presentation/Report	Clear, organized, and visually appealing	Mostly clear and organized	Somewhat disorganized or unclear	Disorganized or incomplete
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V. Synthesis:

How did you arrive at your answers?

How did you find the activity? Can you share and describe your experience?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	6.1	Date:	
Lesson Title/ Topic:	Data and Its Levels of Measurement		
Name:		Grade & Section:	

I. Activity Title: Scenario Worksheet (*Developing Mastery*)

II. Objective:

At the end of the activity, you should be able to determine whether the data presented are qualitative or quantitative and identify its corresponding level of measurement (nominal, ordinal, interval, or ratio).

III. Materials Needed:

pen and paper

IV. Instructions:

Carefully read each scenario provided on the table below. For every situation, determine whether the data presented are qualitative or quantitative, and identify its corresponding level of measurement: nominal, ordinal, interval, or ratio. After identifying, briefly explain your reasoning in the last column.

Scenario	Type of Data	Level of Measurement	Reason
1. Ranking of top 5 students based on academic performance			
2. Daily temperature recorded in Celsius for a week from PAGASA			
3. Number of hours spent on social media per day			
4. Preferred learning modality (modular, online, F2F, blended) of students			
5. Amount of rice consumed by a household per week (in kg)			
6. The modes of transportation students use to go to school (e.g., tricycle, jeep, walking, bicycle)			
7. Monthly electricity consumption of your household in kilowatt-hours.			
8. The type of music students listen to while studying.			
9. Number of correct items the student gets in a quiz.			
Students' satisfaction with school library using a 5-point scale with 1 as "Very Dissatisfied" to 5 as "Very Satisfied."			

V. Synthesis:

1. What helped you determine whether a piece of data is qualitative or quantitative?

2. How do the four levels of measurement (nominal, ordinal, interval, ratio) differ from one another?

3. Why is it important to identify the correct level of measurement when analyzing data?

4. Can you think of real-life examples where understanding the type and level of data is important (e.g., in medicine, education, business)?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	6.2	Date:	
Lesson Title/ Topic:	Data and Its Levels of Measurement		
Name:		Grade & Section:	

I. Activity Title: Group-Based Data Analyst Simulation (*Developing Mastery*)

II. Objective:

At the end of the activity, you should be able to determine the appropriate measures of central tendency to be used in a problem, reasons of using it, and interpret the result.

III. Materials Needed:

pen and paper

IV. Instructions:

Read the following problems and answer what is asked. Use Excel or Jamovi in the computation process.

1. A science teacher recorded the scores of 30 students in a 100-point test. The scores were:
86, 90, 88, 92, 89, 87, 85, 91, 90, 89, 88, 87, 86, 90, 88, 92, 89, 87, 88, 91, 85, 89, 90, 88, 87,
89, 91, 86, 88, 90

Appropriate Measures of Central Tendency:

Reason:

Computed value:

Interpretation:

2. A public school principal is analyzing the monthly family income (in PHP) of 31 students to understand economic disparities:

Monthly Incomes (in PHP):

6,500;	7,000;	7,500;	8,000;	8,500;	9,000;	9,500;
10,000;	10,500;	11,000;	11,500;	12,000;	12,500;	13,000;
13,500;	14,000;	14,500;	15,000;	15,500;	16,000;	16,500;
17,000;	17,500;	18,000;	18,500;	19,000;	19,500;	20,000;
21,000;	23,000;	120,000				

Appropriate Measures of Central Tendency:

Reason:

Computed value:

Interpretation:

3. A PE teacher records the shoe sizes of 40 students as follows:

7, 8, 8, 9, 8, 7, 8, 9, 10, 8, 7, 9, 8, 8, 9, 7, 8, 9, 10, 8, 8, 7, 8, 9, 8, 9, 10, 8, 7, 8, 8, 9, 7, 8, 9, 10, 8, 7, 8, 9

Appropriate Measures of Central Tendency:

Reason:

Computed value:

Interpretation:

V. Synthesis:

1. What helped you determine the appropriate measures of central tendency to be used in a problem?

2. Why is it important to know which measure of central tendency to use in each situation?

3. What skills (e.g., computation, reasoning, interpretation) did you develop during the activity?

4. How can you apply what you've learned in this activity to other subjects or real-life tasks?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	2
Lesson No.:	6.3	Date:	
Lesson Title/ Topic:	Measures of Variability		
Name:		Grade & Section:	

Activity No. 1: Solving Real-Life Problems Using Measures of Variability

Objective:

At the end of the activity, you should be able to compute measures of variability and use the measures to draw conclusions.

Materials Needed:

pen and paper

Instructions:

Analyze and solve the following problems.

A teacher wants to determine and analyze the variability of daily transportation costs (in pesos) of learners since they travel to school using different modes of transportation. She collected the data from 30 learner-commuters to identify the patterns and trends. The data are given below.

55	43	105	34	80	50	55	56	59	49
25	32	60	55	90	29	30	45	57	59
34	36	45	41	47	40	40	45	50	55

a. Compute for the following measures of variability:

(i) range

(ii) interquartile range

(iii) variance

(iv) standard deviation.

b. Interpret the results.

c. What measure of variability is most suitable or appropriate for the given situation? Justify your answer.

A group of researchers surveyed 50 Grade 11 learners to analyze variability in learners' sleep duration (in hours). They divide learners into two groups (i.e. Group 1 – learners who study in the early evening and Group 2 – learners who study late at night. The following data were collected.

Group 1				
5	6.5	9	8	7.5
6	6	7	7	8
5.5	5.5	6.5	7	5.5
4	5.5	7.5	6.5	8.5
8	4.5	5	7.5	9.5

Group 2				
4	7	7	4.5	5.5
5	5.5	6	6.5	6
5	6	7	7.5	5.5
4.5	6	5	5.5	7.5
6	5	4.5	4.5	4.5

For each group, compute for the following:

(i) range

(ii) interquartile range

(iii) variance

(iv) standard deviation.

b. Interpret the results.

c. Answer the following questions:

(i) How does the variability in sleep duration compare between the groups?

(ii) Which group appears to have more stable sleep durations?

(iii) Which group has more extreme fluctuations in sleep hours?

V. Synthesis:

1. How did the activity help you understand measures of variability better?

2. How can you apply what you learned in this activity to real-life scenarios?
